A Practical Guide to Differential privacy machine learning

TOPIC -- DIFFERENTIAL PRIVACY MACHINE LEARNING

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Early research leading to differential privacy In 1977, Tore Dalenius formalized the mathematics of cell suppression. Tore Dalenius was a Swedish statistician who contributed to statistical privacy through his 1977 paper that revealed a key point about statistical databases, which was that databases should not reveal information about an individual that is not otherwise accessible. He also defined a typology for statistical disclosures. In 1979, Dorothy Denning, Peter J. Denning and Mayer D. Schwartz formalized the concept of a Tracker, an adversary that could learn the confidential contents of a statistical database by creating a series of targeted queries and remembering the results. This and future research showed that privacy properties in a database could only be preserved by considering each new query in light of (possibly all) previous queries. This line of work is sometimes called query privacy, with the final result being that tracking the impact of a query on the privacy of individuals in the database was NP-hard.

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The Laplace mechanism adds Laplace noise (i.e. noise from the Laplace distribution, which can be expressed by probability density function noise (y) ∝ exp ■ (− | y | / λ) {\displaystyle {\text{noise}}(y)\propto \exp(-|y|/\lambda)\,\!} , which has mean zero and standard deviation 2 λ {\displaystyle {\sqrt {2}}\lambda \,\!}). Now in our case we define the output function of A {\displaystyle {\mathcal {A}}\,\!} as a real valued function (called as the transcript output by A {\displaystyle {\mathcal {A}}\,\!}) as T A (x) = f (x) + Y {\displaystyle {\mathcal {T}}_{\mathcal {A}}(x)=f(x)+Y\,\!} where Y ∼ Lap (λ) {\displaystyle Y\sim {\text{Lap}}(\lambda)\,\!\,\!} and f {\displaystyle f\,\!} is the original real valued query/function we planned to execute on the database. Now clearly T A (x) {\displaystyle {\mathcal {T}}_{\mathcal {A}}(x)\,\!} can be considered to be a continuous random variable, where

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2008: U.S. Census Bureau, for showing commuting patterns. 2014: Google's RAPPOR, for telemetry such as learning statistics about unwanted software hijacking users' settings. 2015: Google, for sharing historical traffic statistics. 2016: Apple iOS 10, for use in Intelligent personal assistant technology. 2017: Microsoft, for telemetry in Windows. 2020: Social Science One and Facebook, a

55 trillion cell dataset for researchers to learn about elections and democracy. 2021: The US Census Bureau uses differential privacy to release redistricting data from the 2020 Census.

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Sequential composition. If we query an ε-differential privacy mechanism t {\displaystyle t} times, and the randomization of the mechanism is independent for each query, then the result would be ε t {\displaystyle \varepsilon t} -differentially private. In the more general case, if there are n {\displaystyle n} independent mechanisms: M 1 , … , M n {\displaystyle {\mathcal {M}}_{1},\dots ,{\mathcal {M}}_{n}} , whose privacy guarantees are ε 1 , … , ε n {\displaystyle \varepsilon _{1},\dots ,\varepsilon _{n}} differential privacy, respectively, then any function g {\displaystyle g} of them: g (M 1 , … , M n) {\displaystyle g({\mathcal {M}}_{1},\dots ,{\mathcal {M}}_{n})} is (∑ i = 1 n ε i) {\displaystyle \left(\sum \limits _{i=1}^{n}\varepsilon _{i}\right)} -differentially private. Parallel composition. If the previous mechanisms are computed on disjoint subsets of the private database then the function g {\displaystyle g} would be (max i ε i) {\displaystyle (\max _{i}\varepsilon _{i})} -differentially private instead. The other important property for modular use of differential privacy is robustness to post processing. This is defined to mean that for any deterministic or randomized function F {\displaystyle F} defined over the image of the mechanism A {\displaystyle {\mathcal {A}}} , if A {\displaystyle {\mathcal {A}}} satisfies ε-differential privacy, so does F (A) {\displaystyle F({\mathcal {A}})} . The property of composition permits modular construction and analysis of differentially private mechanisms and motivates the concept of the privacy loss budget. If all elements that access sensitive data of a complex mechanisms are separately differentially private, so will be their combination, followed by arbitrary post-processing.

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p d f (T A , D 1 (x) = t) p d f (T A , D 2 (x) = t) = noise (t – f (D 1)) noise (t – f (D 2)) {\displaystyle {\frac {\mathrm {pdf} ({\mathcal {T}}_{{\mathcal {A}},D_{1}}(x)=t)}{\mathrm {pdf} ({\mathcal {T}}_{{\mathcal {A}},D_{2}}(x)=t)}}={\frac {{\text{noise}}(t-f(D_{1}))}{{\text{noise}}(t-f(D_{2}))}}\,\!}

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Stable transformations A transformation T {\displaystyle T} is c {\displaystyle c} -stable if the Hamming distance between T (A) {\displaystyle T(A)} and T (B) {\displaystyle T(B)} is at most c {\displaystyle c} -times the Hamming distance between A {\displaystyle A} and B {\displaystyle B} for any two databases A , B {\displaystyle A,B} . If there is a mechanism M {\displaystyle M} that is ε

{\displaystyle \varepsilon } -differentially private, then the composite mechanism M ■ T {\displaystyle M\circ T} is (ε × c) {\displaystyle (\varepsilon \times c)} -differentially private. This could be generalized to group privacy, as the group size could be thought of as the Hamming distance h {\displaystyle h} between

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Example According to this definition, differential privacy is a condition on the release mechanism (i.e., the trusted party releasing information about the dataset) and not on the dataset itself. Intuitively, this means that for any two datasets that are similar, a given differentially private algorithm will behave approximately the same on both datasets. The definition gives a strong guarantee that presence or absence of an individual will not affect the final output of the algorithm significantly. For example, assume we have a database of medical records D 1 {\displaystyle D_{1}} where each record is a pair (Name, X), where X {\displaystyle X} is a Boolean denoting whether a person has diabetes or not. For example:

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Now suppose a malicious user (often termed an adversary) wants to find whether Chandler has diabetes or not. Suppose he also knows in which row of the database Chandler resides. Now suppose the adversary is only allowed to use a particular form of query Q i {\displaystyle Q_{i}} that returns the partial sum of the first i {\displaystyle i} rows of column X {\displaystyle X} in the database. In order to find Chandler's diabetes status the adversary executes Q 5 (D 1) {\displaystyle Q_{5}(D_{1})} and Q 4 (D 1) {\displaystyle Q_{4}(D_{1})} , then computes their difference. In this example, Q 5 (D 1) = 3 {\displaystyle Q_{5}(D_{1})=3} and Q 4 (D 1) = 2 {\displaystyle Q_{4}(D_{1})=2} , so their difference is 1. This indicates that the "Has Diabetes" field in Chandler's row must be 1. This example highlights how individual information can be compromised even without explicitly querying for the information of a specific individual. Continuing this example, if we construct D 2 {\displaystyle D_{2}} by replacing (Chandler, 1) with (Chandler, 0) then this malicious adversary will be able to distinguish D 2 {\displaystyle D_{2}} from D 1 {\displaystyle D_{1}} by computing Q 5 − Q 4 {\displaystyle Q_{5}-Q_{4}} for each dataset. If the adversary were required to receive the values Q i {\displaystyle Q_{i}} via an ε {\displaystyle \varepsilon } -differentially private algorithm, for a sufficiently small ε {\displaystyle \varepsilon } , then he or she would be unable to distinguish between the two datasets.